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Chapter 6 — L1D/L2 Microarchitecture: FEQ, Snoop Logic, Register Slices, and Timing Behavior (Based on ARM64 Cortex-A72)

6.1 Scope

This chapter describes the microarchitectural structures inside Cortex-A72 that govern cache behavior, timing, and coherency. Although these structures are not architecturally visible, they directly influence the latency and ordering of cache maintenance operations.

Covered topics:

- L1D and L2 internal organization
- FEQ (Fill/Evict Queue)

- Snoop logic and duplicate directory
- Register slices for large L2 configurations
- Writeback pipelines
- Timing behavior relevant to DC CVAU/CVAC/CIVAC
- Interaction with DSB/ISB

These details explain *why* architectural barriers are required.

6.2 L1D Microarchitecture (Cortex-A72)

6.2.1 Structure

- 32 KB
- 64-byte line size
- 16-way set associative
- Write-back
- Virtually indexed, physically tagged (VIPT)
- Non-inclusive relative to L2

6.2.2 Dirty Line Handling

Dirty lines are stored in L1D until:

- Eviction
- DC CVAU
- DC CVAC
- DC CIVAC

L1D does not write back dirty lines spontaneously.

6.2.3 Snoop Behavior

L1D is snooped by L2 using a **duplicate directory**:

- L2 tracks which lines may be dirty in L1D
- Snoop requests query L1D for dirty data
- Snoop responses propagate through FEQ

This mechanism ensures correctness for DC CVAC/CIVAC.

6.3 L2 Microarchitecture (Cortex-A72)

6.3.1 Structure

- 1 MB (typical)
- 64-byte line size
- 16-way set associative
- Write-back
- Non-inclusive, non-exclusive (NINE)

6.3.2 Unified Cache

L2 is unified:

- Holds both instruction and data lines
- Serves as PoU (Point of Unification)
- Serves as the source for instruction refetch after IC IVAU

6.3.3 Dirty Line Handling

Dirty lines may exist in L2. They must be written back to PoC using DC CVAC or DC CIVAC.

6.4 FEQ — Fill/Evict Queue

6.4.1 Purpose

FEQ pipelines cache fill and eviction operations:

- Writebacks
- Snoop responses
- Evictions from L1D/L2
- Line fills from PoC

6.4.2 Behavior

FEQ is asynchronous:

- DC operations enqueue entries
- Writebacks occur over multiple cycles
- Snoop responses may be delayed
- Completion is not guaranteed until **DSB ISH**

6.4.3 Architectural Impact

FEQ is the reason:

- DC CVAC/CIVAC are not instantaneous
- DSB is required after DC operations
- Cache maintenance cannot rely on timing assumptions

6.5 Snoop Logic and Duplicate Directory

6.5.1 Duplicate Directory

L2 maintains a directory of lines that may be dirty in L1D.

6.5.2 Snoop Requests

When DC CVAC/CIVAC is executed:

1. L2 issues snoop requests to L1D
2. L1D returns dirty data if present
3. L2 merges responses
4. Writeback to PoC occurs
5. FEQ pipelines the operation

6.5.3 Architectural Impact

Snoop completion is guaranteed only after **DSB ISH**.

6.6 Register Slices for Large L2 Configurations

6.6.1 Purpose

Register slices are pipeline registers inserted into long internal wires to:

- Improve timing closure
- Support larger L2 sizes
- Support higher frequencies

6.6.2 Behavior

Register slices:

- Add pipeline stages
- Increase internal latency
- Do not change architectural behavior
- Are fully hidden behind DSB/ISB

6.6.3 Architectural Impact

None. Software does not observe register slices.

6.7 Writeback Pipeline

6.7.1 Stages

Writeback from L1D/L2 to PoC typically involves:

1. Snoop request
2. Snoop response
3. Merge dirty data
4. Writeback scheduling
5. Writeback execution
6. Completion acknowledgment

6.7.2 Timing

Writebacks may take multiple cycles due to:

- FEQ depth
- Snoop latency
- Register slice pipeline stages
- Interconnect congestion

6.7.3 Architectural Impact

DSB ISH ensures writeback completion.

6.8 Interaction with DC CVAU/CVAC/CIVAC

6.8.1 DC CVAU

- Pushes dirty data to PoU
- Requires snoops
- Does not guarantee visibility to DMA
- Completion requires DSB ISH

6.8.2 DC CVAC

- Pushes dirty data to PoC
- Requires snoops
- Writeback occurs through FEQ
- Completion requires DSB ISH

6.8.3 DC CIVAC

- Pushes dirty data to PoC
- Invalidates local copies
- Requires snoops
- Completion requires DSB ISH

6.9 Interaction with ISB

ISB does **not** affect data visibility.

ISB affects:

- Instruction pipeline
- BTB
- Return stack
- Prefetch buffers

ISB is irrelevant to data path timing.

6.10 Summary of Microarchitectural Guarantees

Component	Behavior	Architectural Guarantee
L1D	Dirty lines, snooped	DSB ensures visibility
L2	Unified, NINE	PoU source for refetch
FEQ	Pipelines writebacks	DSB drains FEQ
Snoop logic	Queries L1D	DSB ensures completion
Register slices	Timing only	Invisible to software
Writeback pipeline	Multi-cycle	DSB ensures completion

6.11 Exercises (Technical Reference)

Exercise 6.1 — FEQ Behavior

Explain why FEQ requires DSB ISH after DC CVAC/CIVAC.

Exercise 6.2 — Snoop Logic

Describe how duplicate directory snoops ensure correctness for DC CVAC.

Exercise 6.3 — Register Slices

Explain why register slices do not affect architectural behavior.

Exercise 6.4 — Writeback Pipeline

Describe the stages involved in writing back a dirty line to PoC.

Exercise 6.5 — L1D/L2 Interaction

Explain why dirty lines may exist in both L1D and L2.